

Data Cleaning and Analytics

Topic 1 Introduction *-- Week 1*

Dr. Igor Razgon
igor.razgon@durham.ac.uk

When and How?

- Term 1 Michaelmas
- Lectures, reading materials, F2F interaction
- Assessment: evaluated 100% via coursework
- Coursework:
 - Benchtest
 - to be carried out during the last practical (Week 11)

More about the benchtest

- A question bank of 60 **multiple choice** questions
- There are 5 categories of questions (12 questions in each category):
 - Basic Numpy
 - Basic Pandas
 - Data Cleaning
 - Data Analytics
 - Visualization
- Each student will receive 20 questions, 4 questions randomly chosen from each category.
- There will be **one hour** for answering the questions.
- The test will be **open book**: use of any literature of Internet will be allowed (short of consulting with each other).

What is Data Cleaning?

Transformation of the data collected from customers to get rid of:

- **Missing values:**

salary of employees is collected over several months.

But for some employees the information is not available for some months.

- **Duplicate:**

the data for some employees is entered more than once.

- **Outliers:**

Some employers came and left after several weeks.

Information about their salaries will obfuscate the statistical analysis.

What is Data Analytics?

- **Data classification:**

For example, student marks are in the range 0-100,
but they are often classified into classes
(70-100 first class, 60-69 upper second class and so on).

- **Visualization:**

The data is visualized by various plots
(for example, histogram, barplot, boxplot).

- **Interpretation:**

Verbal conclusions are made based on the data visualization.

Books

- **Joel Grus 'Data Science from Scratch. First principles with Python'**

Chapter 2 provides an excellent introduction to basic Python.

Basic introduction to matplotlib.

Data analytics broadly covered but without the use of Pandas or Seaborn.
- **Wes McKinney, 'Python for Data Analysis Data Wrangling with Pandas, NumPy, and IPython':**

Massive coverage of Numpy and Pandas.

Visualization: matplotlib covered but not Seaborn.
- **Jake VanderPlas 'Python Data Science Handbook'**

Broad coverage of NumPy and Pandas.

The book is more of encyclopedic nature.

Web resources

<https://www.w3schools.com/python/numpy/default.asp>

Excellent and well systematized NumPy tutorial.

<https://www.w3schools.com/python/pandas/default.asp>

Also a Pandas tutorial, but in my opinion, it is less systematized.

Formal descriptions can be found simply by googling.

E.g. googling “*pandas cut*” will bring to this webpage:

<https://pandas.pydata.org/docs/reference/api/pandas.cut.html>

Self-exploration

- Sometimes it is difficult to find an answer to a specific question in a book or in the Web
- In this case, I simply carry out experimentation by writing a small fragment of code that would exhibit the answer I am looking for.
- Finally, feel free to ask the instructor.

Plan for the lectures

Week 6:

Introductory lecture.

Reading from and Writing to files

Week 7:

Basic Numpy

Basic Pandas

Week 8:

Data Cleaning (removal of outliers and missing values and duplicates) in Pandas

Analysis of dependencies and classification.

Week 9:

Visualization of one-dimensional data

Visualization of multidimensional data

Week 10: Revision and test preparation.

Self control questions

- Sheets with self control questions will be posted on the Black Board page of the module starting from the next week.
- The questions will test your understanding of the material and highlight some interesting aspects not considered during the lectures.
- You are strongly encouraged to solve the questions and to send me your solutions for review.
- This will provide a two way feedback. You will see you level of familiarity with the material. I will be able to see possible typical mistakes or patterns misunderstanding and discuss them with the students.
- The questions are **not for marking**.

Data Cleaning and Analytics: informal discussion

1. Definition of data cleansing
2. Criteria for high quality data
3. Stages of production of high quality data

Data cleansing

Data cleansing is the process of detecting and correcting (or removing) corrupt or inaccurate records from a record set, table, or database and refers to identifying incomplete, incorrect, inaccurate or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.^[1]

https://en.wikipedia.org/wiki/Data_cleansing#Data_quality

High-quality data criteria

- **Validity**
- **Accuracy**
- **Completeness**
- **Consistency**
- **Uniformity**

Data quality—(1) Validity

The degree to which the measures conform to defined business rules or constraints.

- Data-Type Constraints:** values in a particular column must be of **a particular datatype**, e.g., boolean, numeric, date, etc.
- Range Constraints:** typically, numbers or dates should fall within **a certain range**.
- Mandatory Constraints:** certain columns **cannot be empty**.
- Unique Constraints:** a field, or a combination of fields, must be **unique across a dataset**.
- Set-Membership constraints:** values of a column come from a set of discrete values, e.g. enum values. For example, a person's gender may be male or female.
- Foreign-key constraints:** as in relational databases, a foreign key column can't have a value that does not exist in the referenced primary key.
- Regular expression patterns:** text fields that have to be in **a certain pattern**. For example, phone numbers may be required to have the pattern (999) 999–9999.
- Cross-field validation:** **certain conditions** that span across multiple fields must hold. For example, a patient's date of discharge from the hospital cannot be earlier than the date of admission.

Data quality—(2) Accuracy

The degree of conformity of a measure to **a standard or a true value**.

- While defining all possible valid values allows invalid values to be easily spotted, it does not mean that they are accurate. A valid street address mightn't actually exist.
- Accuracy is very hard to achieve through data-cleansing in the general case because it requires accessing **an external source of data** that contains the true value. Accuracy has been achieved in some cleansing contexts, notably customer contact data, by using external databases that match up zip codes to geographical locations (city and state) and also help verify that street addresses within these zip codes actually exist.
- Another thing to note is the difference between **accuracy** and **precision**. Saying that you live on the earth is, actually true. But, not precise. Where on the earth?. Saying that you live at a particular street address is more precise.

Data quality—(3) Completeness

The degree to which all required measures are known.

- Incompleteness is almost impossible to fix with data cleansing methodology: one cannot infer facts that were not captured when the data in question was initially recorded.
 - questioning the original source if possible, say **re-interviewing** the subject. But the subject may give a different answer or maybe hard to reach again.
- In the case of systems that insist certain columns should not be empty, one may work around the problem by designating a value that indicates **"unknown"** or **"missing"**, but the supplying of default values does not imply that the data has been made complete.

Data quality—(4) Consistency

The degree to which the data is consistent, within the same data set or across multiple data sets.

- Inconsistency occurs when **two data items in the data set contradict** each other. e.g., a customer is recorded in two different systems as having two different current addresses, and only one of them can be correct.
- Fixing inconsistency is not always possible: it requires a variety of strategies - e.g., deciding which data were recorded **more recently**, which data source is likely to be **most reliable** (the latter knowledge may be specific to a given organization), or simply trying to find the truth by **testing** both data items (e.g., calling up the customer).

Data quality—(5) Uniformity

The degree to which the data is specified using **the same unit of measure**.

- In datasets pooled from different locales, weight may be recorded either in pounds or kilos.
- must be converted to a single measure using an arithmetic transformation.

How to produce high-quality data?

The workflow--an iterative process

- 1. Inspection:** Detect unexpected, incorrect, and inconsistent data.
- 2. Cleaning:** Fix or remove the anomalies discovered.
- 3. Verifying:** After cleaning, the results are inspected to verify correctness.
- 4. Reporting:** A report about the changes made and the quality of the currently stored data is recorded.

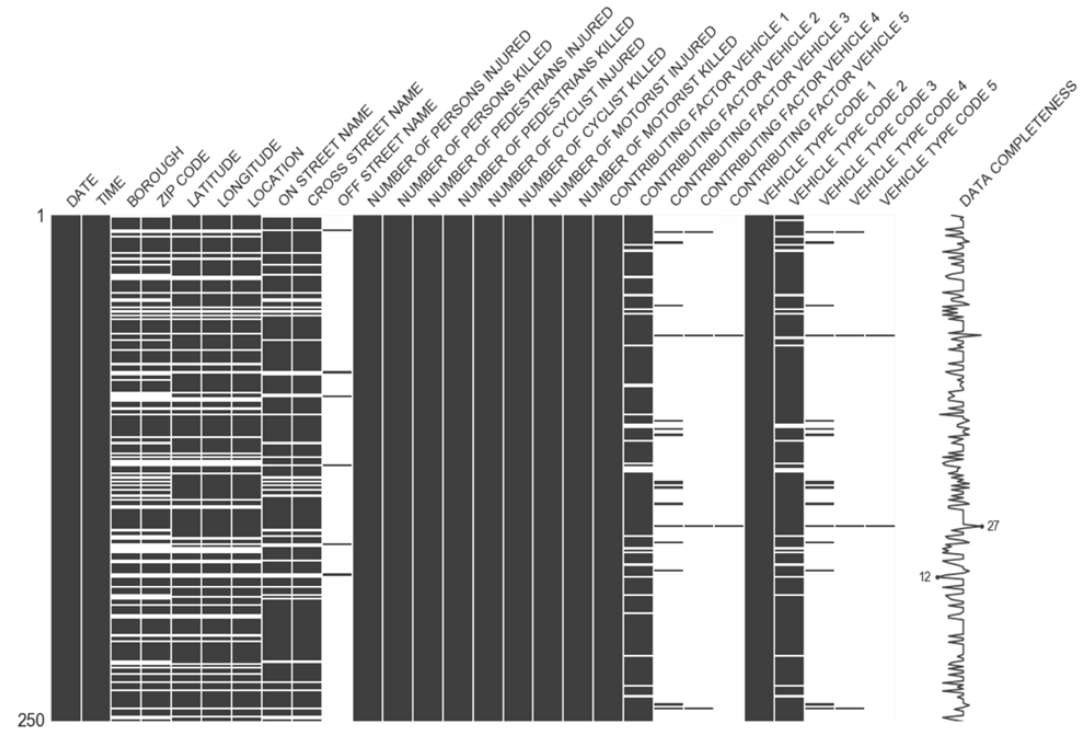
Inspection--exploring the underlying data for error detection

- **Data profiling**: A **summary statistics** about the data, is really helpful to give a general idea about the quality of the data.
 - For example, check whether a particular column conforms to a particular standard or **pattern**. Is the data column recorded as a string or number?
 - How many values are **missing**?. How many unique values in a column, and their **distribution**?. Is this data set is linked to or have a **relationship** with another?

Inspection--exploring the underlying data for error detection

➤ Visualizations

By analyzing and visualizing the data using statistical methods such as mean, standard deviation, range, or quantiles, one can find values that are **unexpected** and thus **erroneous**.



Nullity matrix (Aleksey Bilogur)

<https://github.com/ResidentMario/missingno>

Cleaning--Duplicates

data points that are repeated in your dataset

often happens when for example

- Data are combined from different sources
- The user may hit submit button twice thinking the form wasn't actually submitted.
- A request to online booking was submitted twice correcting wrong information that was entered accidentally in the first time.
- A common symptom is when two users have the same identity number. Or, the same article was scrapped twice. And therefore, they simply should be removed.

News about Ransomware

Cleaning--Type conversion

numbers → numerical data types.

dates → date objects/Unix timestamp (number of seconds)

Categorical values ↔ numbers

Note: The values that can't be converted to the specified type should be converted to NA value (or any), with a warning being displayed. This indicates the value is incorrect and must be fixed.

Cleaning--Standardize

not only recognize the typos

but also put each value in the same standardized format.

- For **strings**, make sure all values are either in lower or upper case.
- For **numerical values**, make sure all values have a certain measurement unit.
- For **dates**, the USA version is not the same as the European version. Recording the date as a timestamp (a number of milliseconds) is not the same as recording the date as a date object.

Cleaning--Scaling / Transformation

transform your data so that it fits within a specific scale, such as 0–100 or 0–1.

It can also help in making certain types of data easier to plot.

- For example, we might want to reduce skewness to assist in plotting (when having such many outliers). The most commonly used functions are **log**, **square root**, and **inverse**.

It can also take place on data that has **different measurement units** or **different scales**.

- Student scores on different exams can't be compared if these two exams are on a different scale. In this case, we need re-scale these two scores to take numbers, say, between 0–1. By scaling, we can plot and compare different scores.

Cleaning--Missing values

- ❑ rarely clean and homogeneous.
- ❑ the missing values are unavoidable
- ❑ different data sources may indicate missing data in different ways.

- Generally, two strategies:
 - using a mask that globally indicates missing values,
 - or choosing a sentinel value that indicates a missing entry.
- In Pandas missing data is represented by two value:
 - None: a Python singleton object that is often used for missing data in Python code.
 - NaN (an acronym for Not a Number): a special floating-point value recognized by all systems that use the **standard IEEE floating-point representation**

Cleaning--Missing values

— One. Drop.

If the missing values in a column rarely happen and occur at random → drop observations (rows)

If most of the column's values are missing, and occur at random → drop the whole column

— Two. Impute: calculate the missing value based on other observations

- 1) Using **statistical values** like mean, median. However, none of these guarantees unbiased data, especially if there are many missing values.
- 2) Using a **linear regression**. Based on the existing data, one can calculate the **best fit** line between two variables, say, house price vs. size m^2 . It is worth mentioning that linear regression models are sensitive to outliers.
- 3) **Hot-deck**: Copying values from other similar records. This is only useful if you have enough available data. And, it can be applied to numerical and categorical data.

Cleaning--Missing values $\neq$ default values $\neq$ “unknown”

— One. Drop.

If the missing values in a column rarely happen and occur at random $\rightarrow$ drop observations (rows)

If most of the column's values are missing, and occur at random $\rightarrow$ drop the whole column

— **Two. Impute:** calculate the missing value based on other observations

— **Three. Flag:** reinforcing the pattern already exist by other features.

What if the data is missing is informative in itself?

when the missing data doesn't happen at random. Take for example a conducted survey where most people from a specific race refuse to answer a certain question.

Cleaning--Outliers

Interquartile range (IQR).

Given a list of values, sort, and then split them into 4 quarters. The range from the end of Q1 till start of Q3 (the middle 50% of the data) is the box in the box plot, where the line inside is the median. Let's take an example. Suppose the values of life expectancies column are:

1,2,2,2|2,3,3,3|5,8,10,15|26,30,45,50

To calculate the quartiles and IQR

- * Quartile 1 (Q1) = 25% = $(2+2)/2 = 2 \Rightarrow$ 25% of numbers ≤ 2
- * Quartile 2 (Q2) = 50% = $(3+5)/2 = 4 \Rightarrow$ 50% of numbers ≤ 4 (median)
- * Quartile 3 (Q3) = 75% = $(15+26)/2 = 20.5 \Rightarrow$ 75% of numbers ≤ 20.5
- * IQR = Q3 — Q1 = $20.5 - 2 = 18.5$ (not really helpful on its own, only when comparing different distributions i.e. data variability)

So, we can say that the “middle 50%” of countries, have life expectancies between 2 and 20.5 years old.

Cleaning--Outliers

They are values that are significantly different from all other observations.

- Any data value that lies more than $(1.5 * \text{IQR})$ away from the Q1 and Q3 quartiles is considered an outlier.
- Even weird, **suspicious values** that are unlikely to happen, should not be removed unless there is a good reason for that.
- It is also worth mentioning that **some models**, like linear regression, are very **sensitive to outliers**. In other words, outliers might throw the model off from where most of the data lie.

Cleaning--In-record & cross-datasets errors

two or more values in the same row or across datasets that contradict with each other

- For example, if we have a dataset about the cost of living in cities. The total column must be equivalent to the sum of rent, transport, and food
- Similarly, a child can't be married. An employee's salary can't be less than the calculated taxes.

Verifying

After cleaning the data, verify correctness by re-inspecting the data and making sure it rules and constraints do hold.

- For example, after filling out the missing data, they might violate any of the rules and constraints.

It might involve some manual correction if not possible otherwise.

Reporting

Reporting **how healthy the data is**

software packages or libraries can generate reports of the changes made, which rules were violated, and how many times

In addition to logging the violations, the causes of these errors should be considered. Why did they happen in the first place?.

Distribution before and after data cleaning